

UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

BOEHRINGER INGELHEIM ANIMAL HEALTH USA INC.,
Petitioner,

v.

KANSAS STATE UNIVERSITY RESEARCH FOUNDATION,
Patent Owner.

PGR2020-00076
Patent 10,450,351 B2

Before ERICA A. FRANKLIN, SHERIDAN K. SNEDDEN, and
ROBERT A. POLLOCK, *Administrative Patent Judges*.

FRANKLIN, *Administrative Patent Judge*.

JUDGMENT

Final Written Decision

Determining All Challenged Claims Unpatentable

Denying Petitioner's Motion to Exclude

Denying Patent Owner's Motion to Exclude

35 U.S.C. § 328(a)

I. INTRODUCTION

This is a Final Written Decision in a post-grant review of claims 1–16 of U.S. Patent No. 10,450,351 B2 (Ex. 1001, “the ’351 patent”). We have jurisdiction under 35 U.S.C. § 6 and enter this Decision pursuant to 35 U.S.C. § 328(a) and 37 C.F.R. § 42.3. Based on the record before us, and for the reasons set forth below, we determine that Boehringer Ingelheim Animal Health USA Inc. (“Petitioner”) has shown, by a preponderance of the evidence, that claims 1–16 (“the challenged claims”) are unpatentable. *See* 35 U.S.C. § 326(e).

A. *Procedural History*

Petitioner filed a Petition requesting post-grant review of the challenged claims. Paper 2 (“Pet.”). Kansas State University Research Foundation (“Patent Owner”) did not file a Preliminary Response to the Petition. On February 3, 2021, we instituted trial to determine whether the challenged claims are unpatentable on the grounds raised in the Petition. Paper 7 (“Dec.”).

Patent Owner filed a Patent Owner’s Response on April 28, 2021. Paper 16 (“PO Resp.”). Petitioner filed a Reply to the Patent Owner’s Response on July 26, 2021. Paper 22 (“Pet. Reply”). Patent Owner filed a Sur-Reply on September 7, 2021. Paper 27 (“PO Sur-Reply”). Patent Owner and Petitioner each filed a motion to exclude evidence and an opposition to each other’s motion. Papers 32–35.

On November 3, 2021, the parties presented arguments at an oral hearing. Paper 31. The transcript of the oral hearing has been entered in the record. Paper 41.

B. Real Parties in Interest

Petitioner identifies itself as the real party in interest. Pet. 1. Patent Owner identifies its real parties in interest as itself, Merck & Co., Inc., Merck Sharp & Dohme Corp., Intervet, Inc., and Intervet International BV. Paper 4, 2.

C. Related Matters

The parties have not provided notice of any active related matters pending in any district court. *See* Pet. 2; Paper 4, 2. Petitioner, however, provides notice of the following related *inter partes* reviews: IPR2018-01788, IPR2018-01789, IPR2019-00374, IPR2019-00381, and IPR2018-00919. Pet. 2. Petitioner also notes the pending prosecution of Application No. 16/600,067, a division of Application No. 15/768,356 from which the '351 patent issued. *Id.*

D. The '351 Patent

1. Eligibility for Post-Grant Review

As set forth in the Institution Decision, we have determined that the '351 patent is eligible for post-grant review. *See* Dec. 3. Set forth below is our analysis of that matter, as presented in the Institution Decision.

Post-grant review is available only for patents “described in section 3(n)(1)” of the Leahy-Smith America Invents Act (“AIA”), Pub L. No. 112-29, 125 Stat. 284 (2011). AIA § 6(f)(2)(A) (2011). Those are patents that issue from applications “that contain[] or contained at any time . . . a claim to a claimed invention that has an effective filing date in section 100(i) of title 35, United States Code, that is on or after” “the expiration of the 18-month period beginning on the date of the enactment of” the AIA. AIA § 3(n)(1). The AIA was enacted on September 16, 2011, therefore, post-grant review is available only for patents that, at any point, contained at least one claim with an effective filing date, as defined by 35 U.S.C. § 100(i), on or after March 16, 2013. The earliest possible filing date for the

'351 patent is October 16, 2015, which falls after the March 16, 2013 date. *See* Ex. 1001, 1:6–8.

The AIA also requires the petition for post-grant review to be filed within nine months of the issue date of the challenged patent. 35 U.S.C. § 321(c). The '351 patent issued on October 22, 2019. Ex. 1001, code (45). The Petition has been accorded a filing date of July 22, 2020, Paper 3, which is within the nine-month window. Thus, Petitioner has timely filed the Petition.

Accordingly, we determine that the '351 patent is eligible for post-grant review.

Id.

2. *Patent Specification*

The '351 patent is directed to porcine circovirus type 3 ("PCV3") immunogenic compositions and methods of making and using such compositions. Ex. 1001, Abstract. The Specification explains that another infectious agent, porcine circovirus type 2 ("PCV2"), was previously identified in pigs with postweaning multisystemic wasting syndrome ("PMWS"). *Id.* at 1:19–21. In addition to PMWS, symptoms of PCV2-associated disease in pigs include pneumonia, porcine dermatitis and nephropathy syndrome ("PDNS"), and reproductive failure. *Id.* at 1:26–28. Commercial vaccines have effectively controlled PCV2-associated disease. *Id.* at 1:24–25.

PCV3 is a new species of circovirus that has been identified in sows having clinical symptoms normally associated with PCV2 infection. *Id.* at 1:32–34. DNA from four sows exhibiting those symptoms was subjected to amplification, inverse PCR gel electrophoresis, and Sanger sequencing of overlapping amplicons spanning the complete genome. *Id.* at 1:67–2:7. As a result, a 2,000 base pair PCV3 genome (SEQ ID NO. 1) was determined. *Id.* at 2:5. Genetic analysis identified an open reading frame ("ORF") (ORF1; SEQ ID NO. 3) encoding a predicted 296 amino acid ("aa") protein

(SEQ ID NO. 4). *Id.* at 2:8–10. A second ORF (ORF2; SEQ ID NO. 5) in the opposite orientation encoded a predicted 214 aa protein (SEQ ID NO. 6). *Id.* at 2:19–21. ORF2 is the immunogenic capsid protein of PCV3. *Id.* at 16:30–31. A third ORF (ORF3; SEQ ID NO. 7) encodes a predicted 233 aa protein (SEQ ID NO. 8). *Id.* at 2:26–28.

The Specification describes, in one aspect, a method of producing and/or recovering recombinant PCV3 ORF2 protein, by “1) infecting a number of susceptible cells in culture with a recombinant viral vector encoding a PCV3 protein, 2) expressing PCV3 protein by the recombinant viral vector, 3) recovering the PCV3 protein, and 4) separating cell debris from the expressed PCV3 protein via a separation step.” *Id.* at 4:1–9. Preferably, an inactivation step is included to inactivate the viral vector prior to recovery of PCV3 protein that will be used in an immunogenic or immunological composition such as a vaccine. *Id.* at 4:10–14. The inactivation step “can be performed as step 5) in addition to steps 1-4 described above . . . just before or just after the filtration or separation step.” *Id.* at 4:14–17. A neutralization step may also be included after step 5. *Id.* at 4:23–24.

Example 3 of the '351 patent demonstrates the cloning, expression and purification of the PCV3 capsid protein. *Id.* at 44:30–31. Example 4 describes “the production and in vitro characterization of an anti-PCV3 capsid monoclonal antibody. BALB/c mice were immunized with purified, truncated capsid protein (35–214 aa)” and “inoculated with 50 µg of antigen mixed with Freund’s incomplete adjuvant bi-weekly for a total of eight weeks.” *Id.* at 44:57–64.

The Specification states, “[i]t will be found that the immunogenic compositions comprising recombinant PCV3 ORF protein as provided herewith are very effective in reducing the severity of or incidence of clinical signs associated with PCV3 infections up to and including the prevention of such signs.” *Id.* at 12:12–16.

E. Illustrative Claims

Petitioner challenges claims 1–16. Claims 1 and 12 are illustrative and read as follows:

1. A composition comprising:
at least one porcine circovirus type 3 recombinant protein
selected from the group consisting of ORF1, ORF2, and
ORF3, wherein said protein is encoded by a nucleotide
sequence having at least 90% sequence homology with a
sequence selected from the group consisting of SEQ ID NO. 3,
SEQ ID NO. 5, SEQ ID NO. 7, or wherein said protein has at
least 90% sequence homology with a sequence selected from
the group consisting of SEQ ID NO. 4, SEQ ID NO. 6, SEQ
ID NO. 8, or any combination thereof; and
a veterinary-acceptable carrier selected from the group
consisting of a coating, a stabilizing agent that increases and/or
maintains product shelf-life and/or enhances stability, a
preservative, an antimicrobial [sic] agent, an antifungal agent,
an isotonic agent, an adsorption delaying agent, or any
combination thereof.
12. A method of inducing a[n] immunological response against
porcine circovirus type 3 comprising the steps of:
administering a composition selected from the group
consisting of the composition of claim 1, the nucleic acid of
claim 4, the composition of claim 9, or any combination
thereof to an animal in need thereof.

Ex. 1001, 71:14–31, 74:6–11.

F. Asserted Grounds of Unpatentability

Petitioner challenges the patentability of claims 1–16 of the ’351 patent on the following grounds:

Claims Challenged	35 U.S.C. §	Basis
1–16	112(a)	Lack of Written Description
1–16	112(a)	Lack of Enablement
5, 6	112(b)	Indefiniteness
1–8	101	Ineligible Subject Matter
1–11	101	Lack of Utility

Petitioner also relies on the Declaration (Ex. 1026) and Reply Declaration (Ex. 1039) of Robert M. Nordgren, Ph.D. Patent Owner relies on the Declaration (Ex. 2052) of Harm HogenEsch, D.V.M., Ph.D.

II. ANALYSIS

A. Legal Standards

To satisfy the written description requirement under 35 U.S.C. § 112(a), the specification must “reasonably convey[] to those skilled in the art that the inventor had possession” of the claimed invention as of the filing date. *Ariad Pharms., Inc. v. Eli Lilly & Co.*, 598 F.3d 1336, 1351 (Fed. Cir. 2010) (en banc) (“*Ariad*”). Although the Specification need not recite the claimed invention *in haec verba*, it must do more than simply render the claimed invention obvious. *Id.* at 1352 (citing *Lockwood v. Am. Airlines Inc.*, 107 F.3d 1565, 1572 (Fed. Cir. 1997) (“*Lockwood*”).

“One shows that one is ‘in possession’ of *the invention* by describing *the invention*, with all its claimed limitations, . . . by such descriptive means . . . that fully set forth the claimed invention.” *Lockwood*, 107 F.3d at 1572 (citation omitted). However, when evaluating the adequacy of the disclosure, we may consider “the existing knowledge in the particular field, the extent and content of the prior art, the maturity of the science or technology, [and] the predictability of the aspect at issue.” *Capon v. Eshhar*, 418 F.3d 1349, 1359 (Fed. Cir. 2005), *cited with approval in Ariad*, 598 F.3d at 1352.

“[A]n adequate written description of a claimed genus requires more than a generic statement of an invention’s boundaries.” *Ariad*, 598 F.3d at 1349 (citing *Regents of the Univ. of Calif. v. Eli Lilly & Co.*, 119 F.3d 1559, 1568 (Fed. Cir. 1997) (“*Eli Lilly*”). “[M]erely drawing a fence around the outer limits of a purported genus is not an adequate substitute for describing a variety of materials constituting the genus and showing that one has invented a genus and not just a species.” *Id.* at 1350.

“[A] sufficient description of a genus instead requires the disclosure of either a representative number of species falling within the scope of the genus or structural features common to the members of the genus so that one of skill in the art can ‘visualize or recognize’ the members of the genus.” *Id.* (quoting *Eli Lilly*, 119 F.3d at 1568–1569). Such written description “requires a precise definition, such as by structure, formula, chemical name, physical properties, or other properties, of species falling within the genus sufficient to distinguish the genus from other materials.” *Id.* (citing *Eli Lilly*, 119 F.3d at 1568).

B. Level of Ordinary Skill in the Art

The level of skill in the art is a factual determination that provides a primary guarantee of objectivity in a patentability analysis. *Al-Site Corp. v. VSI Int'l Inc.*, 174 F.3d 1308, 1324 (Fed. Cir. 1999) (citing *Graham v. John Deere Co.*, 383 U.S. 1, 17–18 (1966); *Ryko Mfg. Co. v. Nu-Star, Inc.*, 950 F.2d 714, 718 (Fed. Cir. 1991)).

Petitioner describes a person having ordinary skill in the art (“POSITA”) as:

hav[ing] held a Doctorate in Veterinary Medicine (D.V.M.) or equivalent, or a Ph.D. or equivalent in immunology, vaccinology, virology, molecular biology, animal science, and/or husbandry, or a closely related field. A POSITA would also have practical knowledge of immunology, including how vaccine candidates are first identified and then subsequently developed. The knowledge may come from the POSITA’s own experience, or it may come through research or work collaborations with other experienced individuals in the medical, pharmaceutical, or biotech industry, *e.g.*, as members of a research team or group.

Pet. 12 (citing Ex. 1026 ¶ 32).

At the institution stage, we found Petitioner’s description of the level of ordinary skill in the art sufficiently supported by the current record, and therefore, adopted Petitioner’s description of a person of ordinary skill in the art for purposes of the Institution Decision. Dec. 7–8.

In the Patent Owner’s Response, Patent Owner asserts the following description of the level of ordinary skill in the art:

A POSITA at the time would have a Ph.D. in microbiology, immunology, vaccinology, molecular biology, virology or pharmaceuticals, or a closely related field. A POSITA with a Ph.D. would also have at least 3-5 years of practical experience identifying and characterizing animal viruses and/or developing animal vaccines, including specific experience with molecular cloning, recombinant viral protein expression and purification,

identifying antigens, adjuvants, vaccine formulation and molecular diagnostics. [] A POSITA would have such relevant experience, within 5-10 years of the invention because of the rapidly developing state of the art. *Id.* A D.V.M. with additional practical experience working with animal viruses and/or on animal vaccine development would be equivalent to a Ph.D.

PO Resp. 18 (citing Ex. 2052 ¶ 58). Patent Owner and Dr. HogenEsch explain that their description differs from Petitioner's description by not including a Ph.D. in animal science or animal husbandry because that degree "would not be necessary or sufficient for a POSITA" as the claims are not directed to the raising or care of animal, but instead to recombinant protein and nucleic compositions and methods of using them to induce an immune response. *Id.* at 18–19 (citing Ex. 2052 ¶ 59). Patent Owner notes also that its description differs from that of Petitioner's by requiring that "a POSITA would need relevant training near the time of the invention to be fully knowledgeable of the relevant advances in the art." *Id.* at 19 (citing Ex. 2052 ¶ 60).

As noted by the Drs. Nordgren and HogenEsch, the differences in these descriptions are "relatively minor." *See* Ex. 1038, 238:21–24; Ex. 1039 ¶ 42. Dr. Nordgren explains that even under the higher skill level required by Patent Owner's description of a POSITA, his opinions would not change. *Id.* at 1039 ¶ 42. Based upon our review, the differences in Petitioner's and Patent Owner's proposed definitions do not impact Petitioner's grounds for unpatentability or Patent Owner's responsive arguments. Nor does our consideration of the issues presented turn on which proposed definition is applied. In any event, having considered the evidence and the arguments, we find Patent Owner's rationale for its more narrow description of the level of ordinary skill in the art to be persuasive and

supported by record as a whole. Accordingly, for this Decision, we adopt Patent Owner’s description of a person of ordinary skill in the art, while maintaining that the prior art reflects the appropriate level of skill at the time of the claimed invention. *See Okajima v. Bourdeau*, 261 F.3d 1350, 1355 (Fed. Cir. 2001).

C. *Claim Construction*

We construe the claims “using the same claim construction standard that would be used to construe the claim in a civil action under 35 U.S.C. [§] 282(b).” 37 C.F.R. § 42.200(b). Under that standard, and absent any special definitions, we give claim terms their ordinary and customary meaning, as would be understood by one of ordinary skill in the art and in accordance with the prosecution history pertaining to the patent. *Id.*

1. *“porcine circovirus type 3 recombinant protein”*

Petitioner asserts that the term “porcine circovirus type 3 recombinant protein,” recited in claim 1, should be construed “to cover proteins that are structurally identical to naturally-occurring PVC3 ORF proteins.” Pet. 24. In support of that position, Petitioner refers to the language in claim 1 reciting that the ORF proteins have “at least 90% sequence homology” with the naturally-occurring ORF protein sequences, i.e., SEQ ID Nos. 4, 6, and 8. *Id.* According to Petitioner, because there is no upper limit on the range of sequence homology in the claim or elsewhere in the Specification, the claim encompasses PCV3 ORF proteins having 100% sequence homology with the naturally-occurring ORF proteins, even though synthesized through recombinant means. *Id.* at 24–25 (citing Ex. 1026 ¶¶ 216–218, 228, 231).

In the Institution Decision, we explained,

Any construction of the claim phrase “porcine circovirus type 3 recombinant protein” must recognize that the PCV3

protein is one obtained by genetic recombination. Insofar as Petitioner asserts that the recited proteins encompass recombinant PCV3 ORF proteins having 100% sequence homology with the naturally-occurring PCV3 ORF proteins, we agree because the claim also recites “wherein said protein has at least 90% sequence homology with a sequence selected from the group consisting of SEQ ID NO. 4, SEQ ID NO. 6, SEQ ID NO. 8,” which necessarily includes those proteins having 100% sequence homology. *See* Pet. 24–25. We recognize also, at this stage in the proceeding, that a recombinant PCV3 ORF protein having 100% sequence homology with the naturally-occurring PCV3 ORF protein would be considered by those having ordinary skill in the art to have an amino acid sequence that is structurally identical to the amino acid sequence of the naturally-occurring PCV3 ORF protein. *See* Ex. 1026 ¶¶ 216–219. The claim phrase “porcine circovirus type 3 recombinant protein” broadly encompasses a number of additional recombinant PCV3 ORF proteins based on the percentage sequence homology of the protein. At this stage in the proceeding, however, we do not find it necessary to provide an express construction for the claim phrase.

Dec. 8–9.

In the Patent Owner Response, Patent Owner asserts that Petitioner’s proposed construction is actually a subject matter eligibility argument. PO Resp. 23. According to Patent Owner, “[t]he term ‘porcine circovirus type 3 recombinant protein’ has a plain and ordinary meaning of ‘genetically engineered PCV3 protein.’” *Id.* (citing Ex. 2052 ¶¶ 91–92).

Having considered the positions of both parties, and the evidence of record, we find that for purposes of this Decision, the ordinary and customary meaning of the claim term “porcine circovirus type 3 recombinant protein,” is a PCV3 protein obtained by genetic engineering, and specifically by recombinant technology, wherein the recombinant protein may either have the same structure and same amino acid sequence as

the naturally occurring PCV3 ORF1–3 proteins or may involve some variation therein. *See* Ex. 1026 ¶¶ 217–218; Ex. 1038, 264:2–9, 266:13–267:3).

2. “*nucleotide sequence encoding ORF1, . . . ORF2, and . . . ORF3*”

Petitioner asserts that the phrase “nucleotide sequence encoding ORF1, . . . ORF2, and . . . ORF3” recited in claim 5 is not clear as claim 4, from which claim 5 depends, does not provide an antecedent basis for the term “ORF.” Pet. 25. According to Petitioner, “[t]o the extent that this term may be construed at all, it should be construed broadly to encompass not only SEQ ID Nos. 3, 5, and 7, but also sequences with at least 90% sequence homology.” *Id.* at 26. At the Institution stage, we explained,

We are not persuaded at this stage, particularly absent any argument or response from Patent Owner, that there is any dispute concerning this claim element that requires an express construction from the Board. We will make a determination on whether such an express construction is appropriate based on the record developed at trial.

Dec. 9–10.

In the Patent Owner Response, Patent Owner asserts that “[a] POSITA would have understood ‘the nucleotide sequence encoding [ORF1, ORF2 or ORF3]’ to mean ‘a nucleotide sequence having at least 90% sequence homology with SEQ ID NOS. 3, 5, or 7 or encoding a sequence having at least 90% sequence homology with SEQ ID NOS. 4, 6 or 8.’” PO Resp. 24 (citing Ex. 2052 ¶¶ 93–96). In support of that position, Patent Owner refers to the following Specification description:

[a] preferred PCV3 ORF2 protein is that of SEQ ID NO. 6, but it is understood by those of skill in the art that this sequence, as well as those for ORF1 and ORF3 *could vary by as much as 10% in sequence homology* and still retain the antigenic characteristics that render it useful in immunogenic compositions.

Id. at 25 (quoting Ex. 1001, 8:49–56) (emphasis added by Patent Owner). Patent Owner asserts, according to that Specification description, “the sequences described and identified by SEQ ID Nos. in the patent for PCV3 ORF1, ORF2 and ORF3 are representative sequences, and ‘ORF1,’ ‘ORF2’ and ‘ORF3’ in the patent also refers to those sequences with at least 90% sequence homology.” *Id.* (citing Ex. 2052 ¶ 95; Ex. 2006, 107:9–17). Further, Patent Owner asserts this interpretation is consistent with the language of the other claims. In particular, Patent Owner refers to the limitation in claim 1 reciting that the composition comprises a PCV3 recombinant protein selected from “ORF1, ORF2 and ORF3, wherein said protein is encoded by a nucleotide sequence having at least 90% sequence homology with a sequence selected from the group consisting of SEQ ID NO. 3, SEQ ID NO. 5, SEQ ID NO. 7.” *Id.* (quoting Ex. 1001, 71:14–21).

Having considered the positions of the parties, and the evidence of record, we agree with Patent Owner that the Specification provides clarity and meaning for the use of the terms ORF1, ORF2, and ORF3, which are not set forth or defined in claim 4, from which claim 5 depends, or in claim 5. Petitioner, in its alternative argument regarding the construction of the phrase involving those terms in claim 5, agrees to some extent. *See* Pet. 26. For purposes of this Decision, and for the reasons discussed by Patent Owner, we construe the claim phrase “nucleotide sequence encoding ORF1, . . . ORF2, and . . . ORF3” recited in claim 5 as meaning “a nucleotide sequence having at least 90% sequence homology with SEQ ID NOS. 3, 5, or 7 or encoding a sequence having at least 90% sequence homology with SEQ ID NOS. 4, 6 or 8.”

3. *immunogenicity*

Patent Owner asserts that, although Petitioner does not propose a construction for any claim term relating to immunogenicity, Petitioner relies on an assumption that the composition claims must be “immunogenic” or “immunological” for its written description and enablement challenges.

PO Resp. 19. Patent Owner contends that such an attempt to import immunogenicity into the composition claims 1–11 should be rejected. *Id.*

Based on our review of the record, we find that this issue is best addressed, to the extent necessary, in our following discussion and analysis of the patentability challenges.

D. *Written Description*

Petitioner asserts that claims 1–16 are unpatentable under 35 U.S.C. § 112(a) for failing to satisfy the written description requirement. Pet. 50–68; Pet. Reply 3–14. Patent Owner disagrees. PO Resp. 33–53; PO Sur-Reply 7–16.

1. *Petitioner’s Contentions*

Petitioner asserts that claim 1 covers a broad genus of amino acid sequences homologous to PCV3 ORFs 1–3, however, the Specification “fails to show that the inventor was in possession of the claimed protein genera—especially those that concern ORFs 1 and 3—as of the effective filing date.” Pet. 52. According to Petitioner, “[w]hile the specification discloses a nucleotide sequence that encodes, and an amino acid sequence that comprises, the naturally-occurring PCV3 ORFs 1-3 (Ex. 1001 at 2:8-38), it provides only a wish or plan for a composition that broadly covers the 90% homology range set forth in the patent (Ex. 1026 (Nordgren Dec.) ¶¶ 167–168).” *Id.* (citing *Eli Lilly*, 119 F.3d at 1566).

In particular, Petitioner asserts that the Specification contains no description to show a POSITA that the inventor was in possession of every recombinant protein with at least 90% sequence or amino acid homology with ORFs 1–3. *Id.* at 53. Petitioner contends that, instead of describing that broad genus, the Specification merely discloses: (1) a single, 2,000 based pair genome; (2) genetic analysis on that genome to identify three ORFs; and (3) a theory, without any testing or support, “that any amino acid (or encoding nucleotide) sequence having at least 90% homology with the disclosed ORF amino acid (or encoding nucleotide) sequences would not only be a PCV3 ORF protein, but also could serve as an immunogenic composition.” *Id.*

According to Petitioner, the “at least 90% sequence homology” limitation causes claim 1 to cover, at a minimum, thousands of homologous amino acid sequences, which amount is exponentially increased by the recitation of “any combination thereof” in the claim. *Id.* (citing Ex. 1026 ¶¶ 115, 125–126, 167). Petitioner asserts that the Specification fails to disclose any description explaining “which of those thousands of different proteins the inventor understood to produce the intended result, and which ones of them would have been operative.” *Id.* Petitioner contends that the Specification, instead, “merely provides that the PCV3-derived amino acid sequences ‘could vary by as much as 10% in sequence homology and still retain the[ir] antigenic characteristics.’” *Id.* (quoting Ex. 1001, 8:49–56). In particular, Petitioner asserts that the Specification “provides no experiments or data, or any specific characterization of the PCV3 genome or its corresponding proteins to suggest possession of an immunogenic composition that comprises them along with a veterinary-acceptable carrier.” *Id.* at 54.

Specifically, regarding ORFs 1 and 3, Petitioner contends that the Specification only speculates as to their potential to serve an immunogenic function and that such speculation runs counter to the expectations of a POSITA in view of related circoviruses like Canine CV, for which it was known that ORFs 1 and 3 have no meaningful immunogenic effect. *Id.* at 54-55 (citing Ex. 1026 ¶¶ 101, 108, 119, 172). Further, Petitioner asserts that the Specification provides no example to demonstrate possession of an immunologic composition with an ORF1 or ORF3 protein. *Id.* at 55 (citing Ex. 1026 ¶¶ 74, 103–109). According to Petitioner, in view of the Specification, a POSITA “would not believe that the inventor had expressed or otherwise isolated any of the claimed PCV3 ORF1 and ORF3 proteins, let alone that the inventor had used them in an immunogenic composition, and therefore would not have understood the inventor to possess them.” *Id.* (citing Ex. 1026 ¶ 172).

Regarding the claimed ORF2 protein-based compositions, Petitioner asserts that there is disclosed only one example of a composition used in an animal, but that composition comprised a “truncated capsid protein” instead of the full-length protein, as claimed. *Id.* (citing Ex. 1001, 44:60–61). Petitioner asserts that the Specification, thus, could not have provided representative members of the genus covered by claim 1 since the Specification failed to provide even a single example that falls within the scope of the claim. *Id.* at 56.

Further, Petitioner asserts that the Specification does not demonstrate that the inventor possessed an entire genus because the Specification “does not assess any epitope or specific region of ORF2 that may be responsible for an immunogenic reaction.” *Id.* Petitioner contends, “[w]hatever the sequence and/or structure that is responsible for the immunogenic response,

the '351 Patent does not say. Instead it leaves that critical work for others to do.” *Id.* (citing Ex. 1001, 14:10–34, 21:22–25). According to Petitioner, “[w]ithout providing an understanding of what portion of PCV3 ORF2 may cause an immune response, and without testing more than a single truncated PCV3 ORF2 protein, it cannot be said that the inventor of the '351 Patent was in possession of the entire genus of claimed proteins that are immunogenic.” *Id.* (citing *Ariad*, 598 F.3d at 1350). Petitioner asserts that, “[a]t bottom, the solo inventor did nothing more than sequence a single PCV3 genome,” and then “broadly claim[ed] a composition with any protein (or encoding nucleotide sequence) with at least 90% homology with any of the ORF1-3 proteins,” without conducting any “experiment or study to show what subset of them, if any, actually may induce an immune response in pigs.” *Id.* at 56–57.

Petitioner asserts that claims 2–16 lack written description for reasons similar to those set forth for claim 1. *See* Pet. 58–68.

2. Patent Owner’s Contentions

Patent Owner begins by asserting that the composition claims, i.e., claims 1–11, “do not require ‘immunogenic’ or ‘immunological’ compositions.” PO Resp. 34. In support of that contention, Patent Owner begins by noting that the language of the composition claims do not recite those term. *Id.* at 20. Next, Patent Owner asserts that “the specification makes clear that the claimed compositions may be used for purposes that do not require immunogenicity or inducing an immune response.” *Id.* (citing Ex. 2052 ¶¶ 80–86). On that point, Patent Owner asserts that, unlike the method claims, i.e., claims 12–16, which are directed to inducing an immunological response, “a POSITA would have understood that this is not the only application for composition claims 1-11.” *Id.* at 20–21. Patent

Owner refers to testimony by Petitioner's expert, Dr. Nordgren, that (a) the composition claims do "not yet require or claim immunogenicity;" and (b) the Specification describes myriad non-vaccine uses of the claimed compositions, including, e.g., diagnostics, compound screening and pathogenesis studies. *Id.* (quoting Ex. 2006, 87:5–11; citing Ex. 2006, 60:10–13, 63:13–19, 73:15–22, 74:11–18, 79:11–13, 80:17–21, 84:18–21; Ex. 2052 ¶¶ 80–86).

Patent Owner contends that the prosecution history "further underscores that the composition claims do not contain the term 'immunogenicity' as a limitation," as the Examiner rejected only the (then-pending) method claims, and not the composition claims, based on a determination that "the specification . . . does not reasonably provide enablement for inducing an immune response" *Id.* (quoting Ex. 1002, 305).

Patent Owner asserts that, to the extent the claims do require an immunological response, the Specification expressly defines "immunogenic or immunological composition" broadly as "a composition of matter that comprises at least one antigen which elicits an immunological response in the host of a cellular and/or antibody-mediated immune response to the composition or vaccine of interest." PO Resp. 21–22 (quoting Ex. 1001, 4:28–32). Patent Owner asserts that "a POSITA would have understood that an 'immunological response' refers to any immune response and need not be protective," i.e., it need not rise to the level of that of a vaccine and induce a protective immune response. *Id.* at 22–23 (citing Ex. 1001, 4:28–37; Ex. 2052 ¶ 90; Ex. 2006, 94:14–19).

Referring to the re-direct examination of Dr. Nordgren at his deposition, Patent Owner asserts that Petitioner's position appears to be that "because compositions with immunological activity are not excluded from the claims there must be written description support for such immunological activity." *Id.* at 34 (citing Ex. 2006, 206:18–208:6). In response, Patent Owner asserts that "[n]othing in the law requires that every possible unclaimed use of a composition (here, a protective vaccine) must be described under Section 112." *Id.* Rather, Patent Owner asserts, "[t]he inquiry is whether the specification describes 'the invention, with all its claimed limitations,' and the 'invention is, for purposes of the 'written description inquiry, whatever is now claimed.'" *Id.* (quoting *Lockwood*, 107 F.3d at 1572 (quoting *Vas-Cath Inc. v. Mahurkar*, 935 F.2d 1555, 1563–64 (Fed. Cir. 1991))).

According to Patent Owner, even if the independent composition claims did require immunogenic or immunological compositions, a POSITA would have understood that the inventor, Dr. Hause, was in possession of such compositions that induce an immunological response. *Id.* at 35. In support of that contention, Petitioner asserts that the Specification: (a) states "preparing an immunogenic composition, such as a vaccine, for invoking an immune response against PCV3' using 'express[ed] and recover[ed] PCV3 ORF protein,'" Ex. 1001, 7:19–26; and (b) describes "the use of the claimed nucleic acids in vaccines and immunological compositions," *id.* at 20:51–63, 32:29–43. Pet. 35. Patent Owner asserts also that because viral proteins are foreign to an animal, "a POSITA would have expected such proteins to, at least, elicit antibody production or a T-cell response when administered to an animal." PO Resp. 35 (citing Ex. 1001, 4:28–37; Ex. 2034, 4; Ex. 2052 ¶¶ 55–57, 114, 148). Patent Owner argues that "[t]he same would be true of

related sequences that have at least 90% sequence homology to PCV3 sequences.” *Id.* (citing Ex. 1001, 8:49–56; Ex. 2052 ¶ 115). Further, Patent Owner asserts that the Specification explains that a POSITA would have known that “adjuvants could be added to a composition to boost immunogenicity if it is used for inducing an immune response in an animal.” *Id.* at 35–36 (citing Ex. 1001, 6:9–7:10).

Patent Owner asserts that Petitioner has not provided “any evidence to suggest that proteins or nucleic acids with at least 90% sequence homology to PCV3 ORFs would *not* be immunogenic.” *Id.* at 36. Patent Owner alleges, “[t]o the contrary, related PCV proteins, including all three ORF proteins of PCV2, were known to be immunogenic.” *Id.* at 36.

Additionally, Patent Owner alleges that “the ’351 Patent demonstrates that PCV3 ORF2 protein is immunogenic in Example 4, by eliciting production of antibodies specific to PCV3.” *Id.* at 36 (citing Ex. 1001, 44:60–67; Ex. 2052 ¶¶ 116–117). Although Example 4 involved immunization with a truncated capsid protein, Patent Owner asserts that the truncated protein is “identical to the corresponding segment of full-length protein” and “it elicited antibodies specifically recognizing native (full-length) PCV3 ORF2 protein expressed in mammalian cells.” *Id.* at 39–40.

Patent Owner responds to Petitioner’s allegation that the Specification does not describe “the sequence and/or structure that is responsible for the immunogenic response” of the ORF2 protein, by asserting that, even if the claims require immunogenic compositions, “a detailed characterization of every antigenic sequence or epitope of the protein, which is very small, is not necessary to practice them.” *Id.* at 40 (citing Ex. 2008 3-4; Ex. 2052 ¶ 130). According to Patent Owner, the ’351 patent discloses an immunological response for the claims because it “discloses that the PCV3

ORF2 protein is immunogenic and a POSITA independently would have expected proteins with at least 90% homology to a PCV3 ORF sequence to be immunogenic.” *Id.* at 40 (citing Ex. 1001, 44:55–45:21, 46:48:23–31; Ex. 2052 ¶ 130).

Regarding Petitioner’s argument that the Specification does not sufficiently describe the broad genus claimed, Patent Owner asserts for claim 1 that “the Specification provides adequate description for the full range of claimed proteins.” PO Resp. 42. According to Patent Owner, “[a] POSITA would have understood that the patent fully describes structural features of the claimed proteins (namely at least 90% sequence homology) that distinguish them from other proteins, and that Dr. Hause invented what is claimed.” *Id.* Patent Owner asserts also that “[t]he ’351 Patent discloses and identifies SEQ ID Nos. 3, 5 and 7 as the nucleic acid sequences of PCV3 ORF1, ORF2 and ORF3, respectively, and that the patent identifies SEQ ID Nos. 4, 6 and 8 as the corresponding amino acid sequences for the proteins encoded by those sequences.” *Id.* at 41 (citing Ex. 1001, 7:42–48, 34:9–19; Ex. 2052 ¶ 105). Additionally, Patent Owner asserts that the Specification describes the use of suitable veterinary-acceptable carriers. *Id.* (citing Ex. 1001, 7:42–48, 34:9–19; Ex. 2052 ¶ 105).

Additionally, Patent Owner asserts that “[t]he specification explains, and a POSITA would independently have understood, that the disclosed sequences are representative of PCV3 ORF1, ORF2 and ORF3 and that sequences with at least 90% sequence homology to those disclosed sequences are closely related structurally and functionally.” *Id.* To support that assertion, Patent Owner relies on the following description in the Specification:

A preferred PCV3 ORF 2 protein is that of SEQ ID NO. 6, but it is understood by those of skill in the art that this sequence, as well as those for ORF1 and ORF3 could *vary by as much as 10% sequence homology and still retain the antigenic characteristics that render it useful in immunogenic compositions.*

Id. at 41–42 (quoting Ex. 1001, 8:49–56) (emphasis added by Patent Owner) (citing Ex. 2052 ¶ 106).

Regarding independent claim 4, Patent Owner relies on the same disclosure to assert that “[a] POSITA would have understood that the patent fully describes structural features of the claimed nucleic acids (namely at least 90% sequence homology to specifically identified representative sequences) that distinguish them from other nucleic acids, and that Dr. Hause invented what is claimed.” *Id.* at 42–43.

For independent claim 9, Patent Owner relies on the same disclosures discussed regarding claims 1 and 4 to assert that “[a] POSITA would have understood that the patent fully describes how to arrange and construct the relevant portions of PCV1 and PCV3 genomic material to practice the claims and that Dr. Hause invented what is claimed.” *Id.* at 45. Patent Owner also supports that contention by asserting that the Specification “describes how to arrange and construct the claimed chimeras by replacing a PCV1 ORF with a corresponding ORF from PCV3,” and “further describes claimed chimaera embodiments where the PCV3 ORF2 is swapped into the PCV1 genomic backbone in place of PCV1 ORF2.” *Id.* at 45 (citing Ex. 1001, 15:18–16:10; Ex. 2052 ¶ 167).

For claim 12, Patent Owner asserts that the ’351 patent adequately describes the claimed methods by providing “specific and extensive disclosures for inducing an immunological response, as defined by the patent, against PCV3 in an animal for each of [the protein, nucleic acid, and

chimeric nucleic acid compositions].” *Id.* at 48 (citing Ex. 1001, 11:24–41, 7:42–8:23, 18:46–62, 20:51–63, 32:29–43, 34:9–19, 17:52–18:18, 34:9–19, 36:30–65, 38:35–53, 40:19–63, 41:29–49; Ex. 2052 ¶¶ 180–186).

Further, regarding claim 12’s recitation of administering “the composition of claim 1, the nucleic acid of claim 4, the composition of claim 9, or any combination thereof,” Patent Owner asserts that “the ’351 patent provides written description support to demonstrate to a POSITA that Dr. Hause was in possession of the claimed compositions capable of inducing an immunological response in an animal” for the same reasons and based on the same disclosures discussed above, regarding claims 1, 4, and 9. *Id.* at 48–49.

Patent Owner asserts that each dependent claim also is adequately supported by the Specification, largely for the same reasons and based on the same disclosures relied upon for the independent claims. *See* PO Resp. 49–53. Patent Owner relies upon additional disclosures in the Specification to describe the additional limitations of the dependent claims. *See id.*

3. *Petitioner’s Reply*

In Petitioner’s Reply, Petitioner reiterates its position that the scope of the challenged claims is “indiscernibly broad” and the ’351 patent disclosure is insufficient to support the claimed genus. Pet. Reply 3. In particular, referring to claims 1–3, Petitioner asserts that the genus of independent claim 1 “includes, among other things, a protein with at least 90% sequence homology to SEQ IDs 4 (ORF1), 6 (ORF2), and 8 (ORF3),” however, the Specification “discloses a single species for each of the ORFs 1–3 within a single, naturally-occurring genome,” without “expressly describing any mutation or substitution within those sequences.” *Id.* (citing Ex. 1001, 1:39–

2:31, 3:36–45; Ex. 1038, 152:20–153:14, 165:2–174:16, 198:19–199:6, 261:19–262:7, 278:14–20; Ex. 1039 ¶¶ 92, 134).

According to Petitioner, even a “10% variance of the smallest protein, ORF2 permits innumerable permutations of up to 21 amino acid substitutions anywhere within the sequence. At each substituted location, one amino acid could be substituted for the other nineteen.” *Id.* at 4.

According to Petitioner, based on that variance alone, “there are over 10^{56} —100 septendecillion—combinations.” *Id.* at 4 (citing Ex. 1039 ¶¶ 43–53; Ex. 1038, 178:17–179:4). Petitioner asserts that, “because of how sequence homology is defined, conservative substitutions are *not* counted; thus even more substitutions could be made if some are conservative.” *Id.* (citing Ex. 1038, 165:2–166:6, 169:13–23; Ex. 1039 ¶¶ 54–55). Further, Petitioner asserts that “because the substitutions that count are, by definition, not conservative, the ’351 Patent provides they could ‘significantly’ change the ‘overall functionality’ of the composition.” *Id.* (citing Ex. 1001, 10:66–11:3; Ex. 1038, 154:16–156:5, 166:7–167:8, 168:10–169:12; Ex. 1039 ¶¶ 68–69, 81, 121, 134). Petitioner asserts that “Dr. Hause did not do any test on any substitution or mutation” and the ’351 patent does not describe “common structure among the claimed genus, such as identifying PCV3 protein epitopes.” *Id.* at 4–5.

Petitioner asserts that independent claim 4 “covers not only the single identified natural genome and its ORF1-3 sequences, it also includes, *inter alia*, nucleotide sequences with at least 90% sequence homology to those ORFs.” *Id.* at 5. Petitioner assert that independent claim 9 “covers a chimeric nucleic acid molecule, where the chimera includes an undefined ‘portion’ of a nucleotide sequence with at least 90% sequence homology to PCV3 ORF1-3 sequences, along with an undefined ‘portion’ of PCV1’s

genome.” *Id.* According to Petitioner, “the number of sequences for these claims is at least as large, if not much larger, than the genus of proteins in Claim 1.” *Id.* (citing Ex. 1039 ¶¶ 56–57).

For method claims 12–16, comprising administering a composition within the genera of claims 1, 4, or 9, Petitioner asserts that “[b]ecause Dr. Hause did not possess those genera, he did not possess administering them.” *Id.* at 6 (citing Ex. 1039 ¶ 152).

In the Petitioner’s Reply, Petitioner responds to Patent Owner’s argument that the challenged claims do not require immunogenicity or protection by asserting that the issue is “not whether the claims *require* it, the question instead is whether the full scope of the claims *cover* vaccine compositions, and whether that scope is adequately described by the specification.” *Id.* at 10 (citing PO Resp. 2–3). Petitioner asserts that Patent Owner’s expert, Dr. HogenEsch, admitted that the challenged claims cover vaccine compositions. *Id.* (citing Ex. 1038, 147:8–148:17, 194:19–195:4, 226:11–16). According to Petitioner, because the claimed carriers are “veterinary-acceptable,” it suggests that the compositions of claims 1, 4, and 9 concern vaccines. *Id.* (citing Ex. 1039 ¶ 59). Also, Petitioner asserts that the dependent claims show that these claims cover vaccines. *Id.* For example, Petitioner refers to the use of the compositions of claims 1, 4 or 9 to obtain a protective effect, as recited in claim 15. *Id.* Petitioner also asserts that Dr. HogenEsch admitted that the purpose of adding an immune stimulant in claims 2, 7, 10 and 13 is for a vaccine, and that the only use for the multi-valent claims 3, 8, 11, and 14, comprising a disease causing organism, was in a vaccine. *Id.* (citing Ex. 1038, 190:15–191:1, 231:17–25).

In particular, Petitioner asserts that the homology variance paragraph relied upon by Patent Owner is directed to vaccines, not diagnostics or other uses. *Id.* at 11 (citing Ex. 1039 ¶¶ 72–74). Petitioner notes that disclosure explains not only that a protein could vary by as much as 10% sequence homology “and still *retain* the *antigenic characteristics* that render it useful in immunogenic compositions,” but it further states that:

- (i) the “*antigenic characteristics* of an immunological composition can be, for example *estimated by challenge experiments*” and (ii) “*the antigenic characteristic* of a modified antigen is still *retained*, when the modified antigen confers at least 70%, preferably 80%, more preferably 90% of the *protective immunity* as compared to the PCV3 ORF protein, encoded by the polynucleotide sequence of SEQID NOS. 3, 5, or 7.”

Id. (quoting Ex. 1001, 8:57–64) (emphasis added by Petitioner); Ex. 1039 ¶¶ 72–74).

According to Petitioner, that passage makes clear that the homology variance concerns vaccines, as admitted by Dr. HogenEsch, because “challenge experiments” and “protective immunity” concern only vaccines and not diagnostics. *Id.* (citing Ex. 1038, 299:1–300:7, 301:1–2).

Petitioner asserts that even if the claims cover other uses, such as diagnostics, compound screening, and pathogenesis studies, such uses are not adequately described, and the Specification must still demonstrate that the inventor possessed their full scope. *Id.* at 12–14.

4. Patent Owner’s Sur-Reply

Patent Owner asserts that Petitioner improperly argues that the Specification must describe protective vaccines because vaccines are a potential application of the challenged claims, even though it is undisputed that the claims do not require or recite a protective vaccine application. PO

Sur-Reply 7–8. Patent Owner asserts that argument was improperly introduced as a new theory in the Petitioner’s Reply and is contrary to law. *Id.* at 3–5, 8. According to Patent Owner, the ’351 patent does not need to describe unrecited potential applications, but only what is claimed. *Id.* (citing, e.g., *Lockwood*, 107 F.3d at 1572) (The “invention is, for purposes of the ‘written description’ inquiry, whatever is now claimed.”).

Regarding Petitioner’s arguments that the Specification does not sufficiently describe PCV3 sequences with “at least 90% sequence homology,” Patent Owner urges that the limitation is not overly broad and is supported. *Id.* at 11. In particular, Patent Owner asserts that “[t]he Examiner was surely aware of the number of claimed sequences . . . but nonetheless accepted that sequences with ‘at least 90% sequence homology’ are sufficiently related.” *Id.* (citing Ex. 1002, 302–306; Ex. 2052 ¶¶ 72-73). Additionally, Patent Owner assert that “sequences with as low as 80% identity are classified as the same circovirus species.” *Id.* at 12 (citing PO Resp. 37-38; Ex. 2052 ¶¶ 126, 155–158; Ex. 2007, 4; Ex. 2009, 3; Ex. 2043, 6). Patent Owner asserts also that Petitioner has filed an application, WO ’452,¹ using the same “at least 90% sequence homology” limitation, which demonstrates that it is “commonly accepted in the field, including for PCV sequences.”² *Id.* at 13.

¹ Iyer, et al., WO 2020/206452 A, published Oct. 8, 2020. (Ex. 2001, “WO ’452”).

² Petitioner acknowledges Patent Owner’s assertion that the claims recited in WO ’452 “do not even concern PCV3,” but relate to proteins homologous to a disclosed ORF2 sequence, and are supported by a “more robust disclosure.” PO Sur-Reply 14 and 14 n.8 (quoting Pet. Reply 24).

5. Discussion

Having reviewed the arguments and the evidence, we determine that, based on the record as a whole, Petitioner has demonstrated by a preponderance of the evidence that claims 1–16 are unpatentable for lack of written description.

a) Claims 1–3

Independent claim 1 is directed to a composition comprising at least one PCV3 recombinant protein selected from the group consisting of ORF1, ORF2, and ORF3, wherein the protein is “*encoded by a nucleotide sequence having at least 90% sequence homology*” with a sequence selected from the group consisting of SEQ ID NO. 3, 5, 7, or wherein said protein “*has at least 90% sequence homology*” with a sequence selected from the group consisting of SEQ ID NO. 4 (ORF1), 6 (ORF2), 8 (ORF3), *or any combination thereof*. Ex. 1001, 71:14–25. The composition further comprises one or any combination of the recited veterinary-acceptable carriers. *Id.* at 71:26–31.

Petitioner argues that claim 1 lacks written description for a number of reasons. In particular, we find persuasive Petitioner’s argument that claim 1 recites a broad genus of amino acid sequences homologous to PCV3 ORFs 1–3, but the Specification fails to describe that genus so as to reasonably convey to those skilled in the art that the inventor had possession of the claimed invention as of the filing date. *See* Pet. 52. To begin, Petitioner has demonstrated persuasively, the “at least 90% sequence homology” limitation recited in claim 1 covers, at a minimum, thousands of amino acid sequences homologous to the naturally-occurring PCV3 ORF sequences, and that amount is exponentially increased because the claim also recites that the

composition may comprise “any combination thereof.” Pet. 53; Ex. 1026 ¶¶ 115, 125, 167.

As Dr. Nordgren credibly explains, based on how the ’351 patent defines “sequence homology” and “conservative substitution,”³ “the 90% sequence homology limitation covers a 10% variance of non-conservative substitutions, in addition to any number of the remaining amino acids which may be conservatively substituted.” Ex. 1039 ¶¶ 54, 55. For example, Petitioner and Dr. Nordgren explain that a 10% variance of even the smallest protein, ORF2, “permits innumerable permutations of up to 21 amino acid substitutions anywhere within the sequence.” Pet. Reply 4; Ex. 1039 ¶ 47.

Specifically, Dr. Nordgren testifies,

To determine the number of possible amino acid combinations for up to 21 substitutions, for example, you would calculate 20^{21} and then subtract one. In other words, you would multiple twenty (the number of total amino acids in existence) by itself for the number of possible substitutions (21), which would yield the total number of distinct combinations possible. Then you would subtract one, which represents the original sequence from which substitutions were made.

Ex. 1039 ¶ 47.

³ The Specification explains that “[s]equence homology” . . . refers to a method of determining the relatedness of two sequences. To determine sequence homology, two or more sequences are optimally aligned, and gaps are introduced if necessary.” Ex. 1001, 10:46–49. “[I]n contrast to ‘sequence identity’, conservative amino acid substitutions are counted as a match when determining sequence homology.” *Id.* at 10:50–53. “A ‘conservative substitution’ refers to the substitution of an amino acid residue or nucleotide with another amino acid residue or nucleotide having similar characteristics or properties including size, hydrophobicity, etc. such that the overall functionality does not change significantly.” *Id.* at 10:66–11:3.

According to Dr. Nordgren, performing that calculation for ORF2 yields over 2×10^{27} change combinations. *Id.* ¶ 48.

Dr. Nordgren demonstrates further that number of change combinations does not reflect the full number of possibilities “because different combinations of amino acid locations (different patterns of amino acids) may be changed.” *Id.* ¶ 49. Dr. Nordgren provided a step-by-step calculation for an ORF2 example, and explained, “[t]o get the total number of combinations, you multiply the number of change combinations (2×10^{27}) times the number of pattern combinations (6×10^{28}), which equals approximately 1.3×10^{56} .” *Id.* ¶¶ 50–53. Thus, as Petitioner asserts, based on a 10% variance, “there are over 10^{56} —100 septendecillion—combinations” for ORF2. Pet. Reply 4; Ex. 1039 ¶ 53. Dr. Nordgren explains that “[t]he number of combinations for ORFs 1 and 3 would be greater, because those are of larger size.” Ex. 1039 ¶ 53.

Dr. Nordgren explains that those calculations do not account for the additional substitutions that may be made because, according to the Specification, “sequence homology counts a conservative substitution as a match, meaning for example every amino acid of ORF2 could be substituted and still retain 100% sequence homology with the original sequence, provided the substitutions are conservative.” *Id.* ¶ 54 (citing Ex. 1001, 10:46–65). So, according to Dr. Nordgren, “there are many more possible combinations of amino acids on top of my calculation above.” *Id.* ¶ 55; see Pet. Reply 4 (“because of how sequence homology is defined, conservative substitutions are *not* counted; thus even more substitutions could be made if some are conservative.”).

We credit that detailed testimony by Dr. Nordgren regarding the scope of claim 1 over the testimony of Dr. HogenEsch that “[t]he ‘at least 90% sequence homology’ limitation does not transform the protein composition claims into the broad claims that Dr. Nordgren imagines.” Ex. 2052 ¶ 121. Dr. HogenEsch does not support his contrary view with any calculations or meaningful discussion of substitutions permitted with the “at least 90% sequence homology” limitation. *See id.* ¶¶ 119–126. Instead, Dr. HogenEsch avers generally that “[a] POSITA would have understood that claims 1-3 recited a clearly defined class of proteins with high sequence homology (at least 90%) to sequences disclosed in the patent” and that the claims are not “overly broad.” *Id.* ¶¶ 120, 125. Notably, however, Dr. HogenEsch agreed in his deposition that claim 1 “does not specify where any amino acid substitutions would be,” or “specify which amino acid substitutions could be made,” and that claim 1 “covers any non-conservative amino acid substitutions as well as any substitutions anywhere in the amino acid sequence of an of the ORFs.” Ex. 1038, 178:5–16. Having considered the testimony of Dr. HogenEsch, we do not find that he has credibly rebutted the persuasive testimony of Dr. Nordgren regarding the scope of the claims in view of the “at least 90% sequence homology” limitation.

Having determined that the claimed genus is significantly broad, we next consider whether the Specification adequately describes that claimed genus with more than a generic statement of its boundaries. *See Ariad*, 598 F.3d at 1349. Patent Owner directs us to the disclosures in the Specification identifying SEQ ID Nos. 3, 5, and 7 as the nucleic acid sequences of PCV3 ORF1, ORF2 and ORF3, respectively, and identifying SEQ ID Nos. 4, 6 and 8 as the corresponding amino acid sequences for the proteins encoded by those sequences. PO Resp. 41 (citing Ex. 1001, 7:42–48, 34:9–19; Ex. 2052

¶ 105). Patent Owner asserts that the '351 patent “fully describes structural features of the claimed proteins (namely at least 90% sequence homology) that distinguish them from other proteins, and that Dr. Hause invented what is claimed.” *Id.* at 42. According to Patent Owner, “[t]he specification explains, and a POSITA would independently have understood, that the disclosed sequences are representative of PCV3 ORF1, ORF2 and ORF3 and that sequences with at least 90% sequence homology to those disclosed sequences are closely related structurally and functionally.” *Id.* at 41. The Specification description that Patent Owner relies on for asserting that the sequences with at least 90% sequence homology are “closely related structurally and functionally” to the disclosed PCV3 ORF1, 2, and 3 sequences is the teaching that:

A preferred PCV3 ORF 2 protein is that of SEQ ID NO. 6, but it is understood by those of skill in the art that this sequence, as well as those for ORF1 and ORF3 *could vary by as much as 10% sequence homology and still retain the antigenic characteristics that render it useful in immunogenic compositions.*

Id. at 41–42 (quoting Ex. 1001, 8:49–56) (emphasis added by Patent Owner); Ex. 2052 ¶ 106.

We do not find that description to be adequate. Essentially, the Specification description that the sequences for PCV3 ORF 1, 2, and 3, could vary by as much as 10% sequence homology just rephrases what is claimed in terms of those sequences having at least 90% sequence homology, adding only that antigenic characteristics are retained that render it useful for immunogenic compositions. A claim, however, “does not become more descriptive by its repetition,” *see Ariad*, 598 F.3d at 1350 (quoting *Eli Lilly*, 119 F.3d at 1568), and “a definition of a useful result rather than a definition of what achieves that result” does not sufficiently

distinguish the genus, *see Ariad*, 598 F.3d at 1350 (quoting *Enzo Biochem, Inc. v. Gen-Probe Inc.*, 323 F.3d 956, 969 (Fed. Cir. 2002)).

As noted by Petitioner and Dr. Nordgren, the '351 patent discloses nucleotide and amino acid sequences corresponding only to the naturally-occurring PCV3 ORFs 1–3. PO Resp. 52; Ex. 1026 ¶ 168. While those sequences may be characterized as “reference” sequences, as they reflect sequences with 100% homology with PCV3 ORFs 1–3, Patent Owner has not demonstrated that they are “representative” sequences for the broad range of sequences claimed. We find that the Specification does not reasonably convey that the inventor made or was otherwise in possession of any of the remaining members of the genus with at least 90% sequence or amino acid homology with ORFs 1–3. PO Resp. 53; Ex. 1026 ¶ 168. We find that Dr. Nordgren credibly characterizes the Specification, as follows:

[T]he '351 Patent outlines a preliminary research plan to eventually make and use the broad range of claimed proteins. That is, the inventor provides a number of DNA and primer sequences that could be used to clone homologous DNA sequences into an expression vector, and theorizes that proteins expressed from sequences having 90% sequence homology will be antigenic. Critically, however, the '351 Patent does not disclose which of the thousands of different claimed homologous sequences, if any, the inventor possessed, much less had discovered to be antigenic. Instead, the inventor merely proposes, without any factual support, that PCV3-derived amino acid sequences “could vary by as much as 10% in sequence homology and still retain the[ir] antigenic characteristics.” Ex. 1001 at 8:49-56. A POSITA would not credit that unsupported speculation, and therefore would not believe that the inventors possessed such a wide range of proteins.

Ex. 1026 ¶ 169. We agree with Dr. Nordgren and find his opinion supported by the record. As Petitioner and Dr. Nordgren observe, the '351 patent does not disclose a representative number of species for the large genus covered

by claim 1. *Id.* ¶ 170; Pet. 55–56. Nor does the Specification provide a sufficient description of such a representative species within the genus, Ex. 1026 ¶ 171, i.e., “structural features common to the members of the genus so that one of skill in the art can ‘visualize or recognize’ the members of the genus.” *Ariad*, 598 F.3d at 1350 (quoting *Eli Lilly*, 119 F.3d at 1568–1569).

For example, as noted by Petitioner and Dr. Nordgren, the Specification does not disclose specific amino acid sequences representing a specific targeted epitope, expressly describe any mutation or substitution within the claimed sequences, or provide some other precise definition to distinguish the genus from other materials. Pet. 56; Pet. Reply 3; Ex. 1026 ¶ 171; Ex. 1039 ¶¶ 92, 134. In other words, the Specification does not explain what, if any, structural features exist (e.g., remain) in sequences that vary by as much as 10% that allow them to retain the antigenic characteristics referenced in the Specification. *See* PO Resp. 40–41 (citing Ex. 1001, 8:49–56). As a result, the Specification does not reasonably convey to those skilled in the art any appreciation for which of the myriad sequences that vary by as much as 10% from the reference sequences will retain the antigenic features referenced in the Specification. *See id.*

We agree with Petitioner that, without describing such characteristics referenced by Dr. Nordgren, the Specification “provides only a wish or a plan for a composition that broadly covers the 90% homology range set forth in the patent,” Pet. 52 (citing *Eli Lilly*, 119 F.3d at 1566; Ex. 1026 ¶¶ 167–168), and leaves the critical work for others to do, *id.* at 56. Indeed, we find that the Specification disclosure that the sequences for ORF1–3 could “vary by as much as 10% sequence homology and still retain the antigenic characteristics that render it useful in immunogenic compositions,” serves to “merely draw a fence around the outer limits of a purported genus [which] is

not an adequate substitute for describing a variety of materials constituting the genus and showing that one has invented a genus and not just a species.” *See Ariad*, 598 F.3d at 1350.

We do not find otherwise, based on Patent Owner’s assertions that Petitioner has “its own patents and patent applications featuring near identical ‘at least 90% sequence homology’ claim limitations for PCV2 and PCV3 sequences, without disclosing (let alone testing) thousands of sequences.” PO Resp. 38. We do not analyze the written description for the currently challenged claims based on the specification or claims of other patents or patent applications. It is the Specification of the challenged claims itself that must demonstrate possession.

Nor do we find written description support for the challenged claims by looking to references, such as Grau-Roma,⁴ a post-priority date journal article discussing genotype definition for a different porcine circovirus, PCV2. *See, e.g.*, PO Resp. 37; Ex. 2052 ¶ 121. Patent Owner and Dr. HogenEsch rely on Grau-Roma’s report of 90.2% nucleotide and 86.5% amino acid sequence homology for PCV2 ORF2 proteins as demonstrating that the recited “at least 90% sequence homology” limitation reflects a well-accepted premise that some variation in the sequences of viral proteins may occur without necessarily affecting their function. PO Resp. 37 (citing Ex. 2020, 8; Ex. 2052 ¶¶ 121–124); PO Sur-Reply 12. Yet, neither Patent Owner nor Dr. HogenEsch explains persuasively that a POSITA would have understood that the inventor of the challenged claims, directed to PCV3

⁴ Grau-Roma et al., *A proposal on porcine circovirus type 2 (PCV2) genotype definition and their relation with postweaning multisystemic wasting syndrome (PMWS) occurrence*, 128 VET. MICROBIOLOGY 23–35 (2008) (Ex. 2020, “Grau-Roma”).

recombinant protein compositions, had possession of those claimed inventions based on Grau-Roma's disclosures regarding its study of PCV2 ORF2 sequences and the nucleotide and amino acid homology observed between those sequences. *See* Ex. 2020, 8.

For the same reasons, we do not assign persuasive weight to Dr. HogenEsch's testimony that the "at least 90% sequence homology" limitation is "reasonable because a POSITA would have known that the ORF sequences of different PCV species were less than 90% homologous," based on Allan.⁵ Ex. 2052 ¶ 126. As Dr. HogenEsch acknowledges, Allan describes PCV1 and PCV2 ORF 1 and 2 nucleotide and sequence homology percentages, i.e., "ORF1 of PCV1 and PCV2 were reported to have 83% nucleotide homology and 86% protein sequence homology, while ORF2 of PCV1 and PCV2 were predicted to have 67% nucleotide sequence homology and 65% protein homology." *Id.* Yet, Dr. HogenEsch does not explain or demonstrate how that reported homology relates to ORFs 1, 2 or 3 of PCV3. At most, these extrinsic reports and studies, if presented in prior art, may suggest that the homology limitation in the '351 patent would have been obvious. However, even if incorporated into the Specification, a description that merely renders an invention obvious does not satisfy the written description requirement. *See Lockwood*, 107 F.3d at 1571–1572.

Accordingly, based on the current record, we determine that Petitioner has shown by a preponderance of the evidence that independent claim 1 lacks written description support because a person of ordinary skill

⁵ Allan et al., *Porcine circoviruses: a review*, 12 J. VET. DIAGNOSTICS INVEST. 3–14 (2000) (Ex. 2009, "Allan").

in the art would not have recognized from Specification that the inventor had possession of compositions comprising the broad genus claimed.⁶

Dependent claim 2 recites “[t]he composition of Claim 1, further comprising an immune stimulant.” *See* Ex. 1001, 71:32–33. Claim 3 recites, in part, “[t]he composition of claim 1, further comprising at least one immunological[ly] active component against another disease-causing organism in swine selected from the group” listed in the claim. *See* Ex. 1001, 71:34–62. Because claims 2 and 3 depend from claim 1 and do not further limit or narrow the broad genus of claim 1, we determine that those claims lack adequate written description support for at least the same reasons as claim 1.

b) Claims 4–8

Independent claim 4 and dependent claims 5–8 are directed to “a nucleic acid comprising a nucleotide sequence” selected from a group of sequences derived from the PCV3 genome. *See* Ex. 1001, 71:63–73:6. As Petitioner asserts, these claims cover the single identified natural genome and its ORF1–3 sequences, as well as, nucleotide sequences with at least 90% sequence homology to those ORFs. Pet. Reply 5. Thus, we agree with Petitioner that “[t]he number of sequences for these claims is at least as large, if not much larger, than the genus of proteins in Claim 1.” *Id.* (citing Ex. 1039 ¶¶ 56–57).

⁶ We do not reach Petitioner’s arguments that claim 1 also fails to provide written description support for vaccine compositions and non-vaccine uses of the composition, as we determined that the claims already lack written description because the Specification insufficiently describes the claimed genus based on the “at least 90% sequence homology” limitation.

Based on our review of the arguments and evidence, which largely mirror those presented for claim 1 with respect to the “at least 90% sequence homology” limitation, we find that Petitioner has shown by a preponderance of the evidence that the ’351 patent lacks adequate written description support for the broad genus of nucleic acids in claims 4–8, for reasons similar to those discussed for claim 1, i.e., the Specification does not sufficiently describe representative species or any common features of the genus, beyond teaching that the sequences for PCV3 ORF1–3 could vary by as much as 10% sequence homology.

c) Claims 9–11

Independent claim 9 and dependent claims 10–11 are directed to compositions comprising a chimeric nucleic acid molecule including a plasmid nucleotide sequence containing a portion of PCV1 nucleic acid and a portion of PCV3 nucleic acid, wherein the PCV3 nucleic acid has at least 90% sequence homology with a sequence selected from the group consisting of SEQ ID Nos. 3, 5, 7, or wherein the PCV3 nucleic acid encodes a protein having at least 90% sequence homology with a sequence selected from the group consisting of SEQ ID Nos. 4, 6, 8, or any combination thereof; and a veterinary-acceptable carrier. Ex. 1001, 73:7–74:5.

In terms of the scope, Petitioner asserts that these claims permit “any of [the covered] ORF1-3 nucleic acid sequences to be combined with *any* portion of PCV1 nucleic acid, regardless of length or composition.” Pet. 63 (citing Ex. 1026 ¶¶ 187–192). We agree and find that the claimed genus is extremely broad. According to Petitioner, these claims lack written description for at least the same reasons as claims 1 and 4 because claims 9–11 similarly cover a broad genus of possible nucleic acid combinations between PCV1 and PCV3, while the Specification conveys only to a

POSITA that the inventor disclosed only a wish or a plan for such a composition and not that the inventor was in possession of one. Pet. 62–63; Ex. 1039 ¶ 148. As with claims 1 and 4, Petitioner asserts the Specification does not sufficiently describe representative species or any common features of the genus, beyond teaching that the sequences for PCV ORF1–3 could vary by as much as 10% sequence homology. Pet. Reply 6. Patent Owner relies upon the same arguments and the Specification disclosures here that it relied upon for claims 1 and 4 that to assert that ’351 patent “discloses and identifies representative PCV3 ORF1, ORF2 and ORF3 nucleic acid and amino acid sequences, and sequences with at least 90% homology to those sequences.” PO Resp. 44.

We do not find those arguments persuasive for the same reasons discussed regarding claims 1 and 4. Moreover, we credit Dr. Nordgren’s testimony that the Specification “provides no actual example of a single chimera. The only hypothetical chimeric construct it does provide, which swaps ORF2 of PCV1 or ORF2 of PCV3, is not close in magnitude to the full scope of these chimeric claims. No common structure or function is provided across the various chimera constructs.” Ex. 1039 ¶ 149.

Thus, based on our review of the arguments and evidence, we find that Petitioner has shown by a preponderance of the evidence that the ’351 patent lacks adequate written description support for claims 9–11.

d) Claims 12–16

Claims 12–16 are directed to a method of inducing an immunological response against PCV3 comprising administering a composition selected from the group consisting of the composition of claim 1, the nucleic acid of claim 4, the composition of claim 9, or any combination thereof to an animal in need thereof. Ex. 1001, 74:6–48. Because we have determined that the

Specification does not provide an adequate written description of the composition of claim 1, the nucleic acid of claim 4, or the composition of claim 9, we find that claims 12–16 lack written description for at least the same reasons discussed for those claims.

Thus, based on our review of the arguments and evidence, we find that Petitioner has shown by a preponderance of the evidence that the '351 patent lacks adequate written description support for claims 12–16.

Based on the foregoing, we determine that claims 1–16 are unpatentable under 35 U.S.C. § 112(a) for lack of written description.

E. Remaining Grounds

In the remaining grounds, Petitioner challenges the patentability of claims 1–16 by asserting that: (a) claims 1–8 are ineligible subject matter; (b) claims 1–16 lack enablement; (c) claims 5–6 are indefinite; and (d) claims 1–11 lack utility. Pet. 5. Because we have already determined that all of the challenged claims are unpatentable under 35 U.S.C. § 112(a) for lack of written description, we need not reach Petitioner's remaining grounds that challenge those same claims. *See SAS Inst., Inc. v. Iancu*, 138 S. Ct. 1348, 1359 (2018) (holding that a petitioner "is entitled to a final written decision addressing all of the claims it has challenged").

III. MOTIONS TO EXCLUDE

A. *Patent Owner's Motion to Exclude*

Patent Owner moves to exclude the testimony of Petitioner's expert, Dr. Nordgren, regarding WO '452, in paragraphs 90–94 of his rebuttal declaration, Exhibit 1039. Paper 32. According to Patent Owner, that testimony by Dr. Nordgren is inadmissible under the Federal Rules of Evidence ("FRE"), specifically, FRE 702. *Id.* at 5–6. Patent Owner asserts that Dr. Nordgren's deposition testimony demonstrates that he did not understand the publication and that he did not review it in sufficient detail to know what it disclosed. *Id.* at 6 (citing Ex. 2059, 58:1–61:5). Patent Owner asserts that Dr. Nordgren's opinions regarding WO '452 necessitate exclusion, as opposed to being assigned little or no weight. *Id.* at 7.

Petitioner opposes the motion. Paper 34. According to Petitioner, Dr. Nordgren's testimony regarding WO '452 is limited to rebutting Patent Owner's arguments regarding its purported similarities to the '351 patent. *Id.* at 5. Petitioner notes that Dr. Nordgren explains that he reviewed WO '452 "at a high level" for this matter, but that he did not undertake an effort to make a determination whether the claimed inventions described therein were valid. *Id.* (citing Ex. 1039 ¶ 90).

As the moving party, Patent Owner has the burden of proof to establish that it is entitled to the requested relief. 37 C.F.R. § 42.20(c). Having considered the arguments and evidence, we determine that Patent Owner has not met that burden of proof. Despite Patent Owner's position, WO '452 is not under review in this proceeding. As explained by Petitioner, Dr. Nordgren addressed that application only to an extent that Petitioner and he deemed necessary to rebut contentions made by Patent Owner regarding its relevance in this case. Dr. Nordgren explained that he had performed

only a cursory review of the application. Any consideration by the Board of his opinions regarding WO '452 will be in view of that caveat. However, based on our evaluation, while Dr. Nordgren's review of the application may not have been sufficient to provide an informed response to some of Patent Owner's deposition questions, Dr. Nordgren was candid about that fact. More to the point, we do not find that the challenged testimony in his declaration exceeds the scope of his review and consideration of WO '452 such that it is irrelevant or unreliable.

Accordingly, we deny Patent Owner's Motion to Exclude.

B. Petitioner's Motion to Exclude

Petitioner moves to exclude Exhibits 2001, 2015, 2016, 2031, 2049, and 2055–2058, and portions of the declaration by Petitioner's Expert, Dr. HogenEsch, Exhibit 2052, that rely on those exhibits. Paper 33, 8. According to Petitioner, Exhibits 2001, 2015, 2016, 2031, 2049, and 2055–2058 “post-date the '351 Patent's claimed priority date by years and have no bearing on the time-of-patent claim construction, written description, and enablement inquiries,” and/or concern viruses other than PCV3. *Id.* at 8, 11–12. Petitioner asserts, for those reasons, the exhibits should be excluded under FRE 402 and 403. *Id.* Additionally, Petitioner asserts that Exhibits 2055–2058 should also be excluded as hearsay under FRE 801 and 802, for which no exception applies, to the extent Patent Owner intends to rely on them for the truth of the matter it contains. *Id.* at 16–20.

Patent Owner opposes the motion. Paper 35. Patent Owner contends that the challenged exhibits are relevant and admissible, despite the fact that they represent post-priority date evidence. *Id.* at 1. Patent Owner supports its position by asserting that Exhibit 2001, the WO '452 application, “borrows heavily from the '351 Patent including verbatim disclosures and

strikingly similar claim language,” and that a reasonable factfinder would understand that it contains “relevant admissions regarding the validity of the ’351 Patent.” *Id.* at 7. Patent Owner asserts that Exhibits 2015, 2016, 2031, and 2049 disclose PCV2 vaccines on the market before the filing date of the ’351 patent. *Id.* at 1–2, 7–8. Patent Owner asserts that Ex. 2057 and 2058 rebut Petitioner’s assertion that long nucleic acid probes were useless as of the filing date of the ’351 patent, as the exhibits demonstrate that they continue to be used. *Id.* at 2, 8–9.

Regarding Petitioner’s hearsay challenges to Exhibits 2055–2058, Patent Owner asserts that Petitioner waived those objections by not raising them when Patent Owner introduced the exhibits at Dr. Nordgren’s deposition. *Id.* at 2.

As the moving party, Petitioner has the burden of proof to establish that it is entitled to the requested relief. 37 C.F.R. § 42.20(c). Having considered the arguments and evidence, we determine that Petitioner has not met that burden of proof. Although we find some merit in Petitioner’s motion to exclude under FRE 402 and 403, we do not find that there is sufficient cause to exclude the challenged exhibits in view of our analysis in the Decision regarding them. In particular, any consideration by the Board of the challenged exhibits has taken into account their post-priority date status. Further, as discussed in our analysis, we have recognized and taken into account the exhibits that are directed to PCV2 instead of PCV3. Thus, we do not find that maintaining the exhibits in this proceeding results in any unfair prejudice to Petitioner, or that they have confused the issues before us.

Accordingly, we deny Petitioner’s Motion to Exclude Exhibits 2001, 2015, 2016, 2031, 2049, and 2055–2058, and portions Exhibit 2052 that rely on those exhibits under FRE 402 and 403.

Regarding the hearsay challenges for Exhibits 2055–2058 under FRE 801 and 802, Petitioner asserts that it timely objected to the exhibits. Paper 33, 10. In support of that contention, Petitioner cites to portions of Dr. Nordgren’s deposition, Exhibit 2059, where the objection was made. *Id.* at 16–19. We have reviewed those objections and observe that none were made based on hearsay. *See* Ex. 2059, 74, 80–81, 87–88, 92. Instead, the objections were limited to “beyond the scope of the record,” “relevance,” and “lacking foundation.” *Id.* As Patent Owner correctly asserts, the Patent Trial and Appeal Board Consolidated Trial Practice Guide⁷ (“CTPG”), issued in November 2019, explains that “[a] party wishing to challenge the admissibility of deposition evidence must make an objection during the deposition.” CTPG 78 (citing 37 C.F.R. § 42.64(a)). Because Petitioner’s evidence does not establish that it has done so, the hearsay objections have been waived.

Accordingly, we deny Petitioner’s Motion to Exclude Exhibits 2055–2058 under FRE 801 and 802.

⁷ Available at <https://www.uspto.gov/TrialPracticeGuideConsolidated>.

IV. CONCLUSION⁸

For the foregoing reasons, we conclude that Petitioner has shown by a preponderance of the evidence that claims 1–16 are unpatentable under 35 U.S.C. § 112(a) for lack of written description. The results of this conclusion are summarized in the table following the Order. Further, we deny both parties’ motions to exclude evidence.

V. ORDER

In consideration of the foregoing, it is hereby:

ORDERED that claims 1–16 of U.S. Patent No. 10,450,351 B2 have been shown to be unpatentable;

FURTHER ORDERED that Patent Owner’s Motion to Exclude is denied;

FURTHER ORDERED that Petitioner’s Motion to Exclude is denied; and

FURTHER ORDERED that, because this is a Final Written Decision, any party to this proceeding seeking judicial review of the decision must comply with the notice and service requirements of 37 C.F.R. § 90.2.

⁸ Should Patent Owner wish to pursue amendment of the challenged claims in a reissue or reexamination proceeding subsequent to the issuance of this decision, we draw Patent Owner’s attention to the April 2019 *Notice Regarding Options for Amendments by Patent Owner Through Reissue or Reexamination During a Pending AIA Trial Proceeding*. See 84 Fed. Reg. 16,654 (Apr. 22, 2019). If Patent Owner chooses one of those options, we remind Patent Owner of its continuing obligation to notify the Board of any such related matters in updated mandatory notices. See 37 C.F.R. § 42.8(a)(3), (b)(2).

In summary:

Claims	35 U.S.C. §	Reference(s)/Basis	Claims Shown Unpatentable	Claims Not shown Unpatentable
1–16	112(a)	Lack of Written Description	1–16	
1–16	112(a) ⁹	Enablement		
5, 6	112(b) ¹⁰	Indefiniteness		
1–8	101 ¹¹	Ineligible Subject Matter		
1–11	101 ¹²	Lack of Utility		
Overall Outcome			1–16	

⁹ As discussed above, in Section II.E., we do not reach the enablement ground.

¹⁰ As discussed above, in Section II.E., we do not reach the indefiniteness ground.

¹¹ As discussed above, in Section II.E., we do not reach the ineligible subject matter ground.

¹² As discussed above, in Section II.E., we do not reach the lack of utility ground.

PGR2020-00076
Patent 10,450,351 B2

PETITIONER:

Judy Jarecki-Black
Richard Seeger
BOEHRINGER INGELHEIM ANIMAL HEALTH USA INC.
judy.jarecki@boehringeringelheim.com
richard.seeger@boehringeringelheim.com

Matthew W. Howell
Pamela Holland Councill
ALSTON & BIRD LLP
matthew.howell@alston.com
pamela.councill@alston.com

PATENT OWNER:

Arlene L. Chow
Ernest Yakob
LATHAM & WATKINS LLP
arlene.chow@lw.com
ernest.yakob@lw.com

Scott R. Brown
Crissa A. Cook
HOVEY WILLIAMS LLP
sbrown@hoveywilliams.com
ccook@hoveywilliams.com